Design and Analysis of Algorithms (PCC-CS404)

Course Details

Credit Points: 3

Pre-Requisites

1. Knowledge of data structures and basic programming ability

Syllabus Outline

Unit 1

Characteristics of Algorithm

- Analysis of algorithm: Asymptotic analysis of complexity bounds — best, average and worst-case behaviour; Performance measurements of Algorithm, Time and space trade-offs.
- Analysis of recursive algorithms through recurrence relations: Substitution method, Recursion tree method and Masters' theorem.

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Unit 2

Fundamental Algorithmic Strategies

- Brute-Force, Greedy, Dynamic Programming, Branch-and-Bound and Backtracking methodologies for the design of algorithms; Illustrations of these techniques for Problem-Solving, Bin Packing, Knapsack, TSP. Heuristics — characteristics and their application domains.

Unit 3

Graph and Tree Algorithms

- Traversal algorithms: Depth First Search (DFS) and Breadth First Search (BFS); Shortest path algorithms, Transitive closure, Minimum Spanning Tree, Topological sorting, Network Flow Algorithm.

Unit 4

Tractable and Intractable Problems

- Computability of Algorithms, Computability classes — P, NP, NP-complete and NP-hard. Cook's theorem, Standard NP-complete problems and Reduction techniques.

Unit 5

Advanced Topics

- Approximation algorithms, Randomized algorithms, Class of problems beyond NP — PSPACE.

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Design & Analysis of Algorithms Lab (PCC-CS494)

Credit Points: 2

Laboratory Experiments

Divide and Conquer:

- Implement Binary Search using Divide and Conquer approach
- Implement Merge Sort using Divide and Conquer approach
- Implement Quick Sort using Divide and Conquer approach
- Find Maximum and Minimum element from an array of integers using Divide and Conquer approach
- Find the minimum number of scalar multiplications needed for chain of matrices (Matrix Chain Multiplication)
- Implement all pairs shortest path for a graph (Floyd-Warshall Algorithm) and Traveling Salesman Problem
- Implement Single Source Shortest Path for a graph (Dijkstra, Bellman-Ford Algorithm)

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Branch and Bound:

- Implement 15 Puzzle Problem

Backtracking:

- Implement 8 Queen Problem
- Graph Colouring Problem and Hamiltonian Problem

Greedy Method:

- Knapsack Problem and Job Sequencing with Deadlines
- Minimum Cost Spanning Tree by Prim's Algorithm and Kruskal's Algorithm

Graph Traversal Algorithms:

- Implement Breadth First Search (BFS) and Depth First Search (DFS)